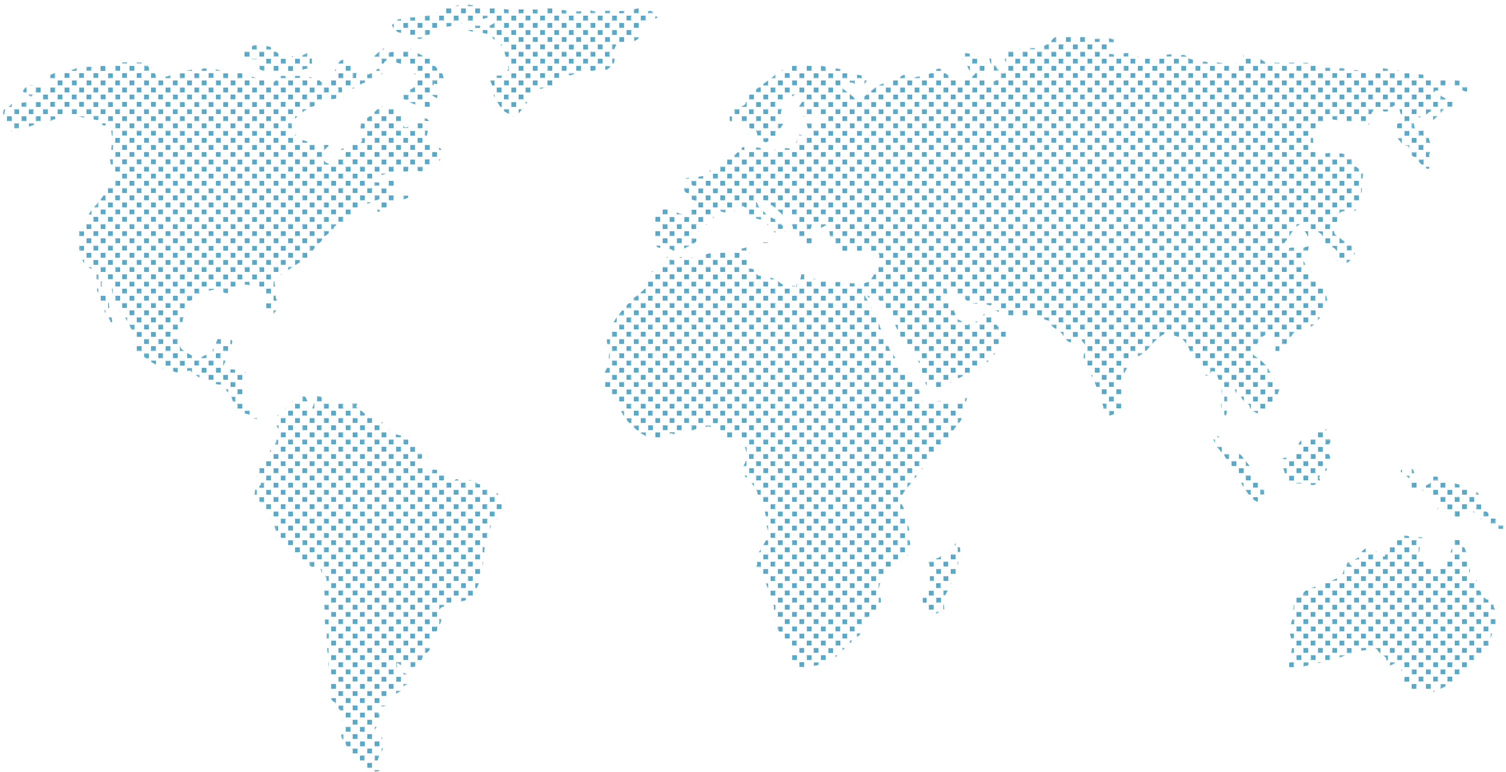

Document Purpose:

Discussion document to describe how text based AI Large Language Models (LLM) such as ChatGPT work and outline potential benefits and costs of getting started with self-hosted and third-party AI in 2025.

Chris Tyler – September 2025

Version	Issued By	Date	Comments
1.0	Chris Tyler	22nd Sept 2025	Document Issued.

1. Overview

The aim is to explain at 'high level' using non-technical language, how text based AI Large Language Models work (see Section 2) and outline some potential benefits and costs of self-hosted and third-party AI in 2025.

AI Large Language Models (LLMs) in 2025, can be viewed as 'Information Processing Toolkits'. These 'tools' can rapidly find, analyse, summarise, translate and generate information, helping their users to work with complex data, answer questions and create content efficiently. This makes otherwise unstructured information more accessible and actionable (see Section 3 for some examples).

ChatGPT, Google Gemini, Microsoft Copilot (and others) offer free versions of their online AI LLMs for personal use and are probably already being used by individuals within most organisations. For anyone else, these free versions are a good way to get an insight into current AI LLM capabilities and limitations.

In addition, community AI LLMs derived from ChatGPT, Google Gemini, Microsoft and others, can be freely downloaded from reputable sites such as Ollama Library and Huggingface (see Section 6) and run locally.

Running AI LLMs locally does not need internet or external 3rd party access. Smaller models such as Llama 3.2 from Meta, are performant on most reasonably modern Desktop PCs, Laptops and Servers. Self-hosting offers cost effective, private access to AI Large Language Models (LLMs), as discussed further in Section 4.

2. What are AI Large Language Models (LLMs) and how do they work?

ChatGPT, Google Gemini, Microsoft Copilot et al, are essentially all large pattern-matching systems, that generate plausible, human-like content by being trained to predict the next most likely word in a sentence, similar to a highly sophisticated autocomplete. For a visual explanation, watch this 7 min video: AI Large Language Models Explained Briefly

Large Language Models (LLMs) are trained on vast amounts of text from books and the internet, allowing them to learn grammar, context and factual patterns.

To strengthen their position and despite the vast costs of developing and training their products, AI companies such as Meta, Google, Open AI (ChatGPT) and others, release free to use versions of their LLMs, to drive widespread community adoption of their models and strengthen their position, by encouraging developers to build on their platforms.

Thus, community versions of ChatGPT, Google Gemini and Microsoft Copilot are free for personal use, without needing an account or logging in. These are very usable despite not being the latest models. Subscriptions to the latest online models with more features, typically cost between £10 and £20 per month per user.

Although LLM training and enterprise LLM models are enormous, perfectly usable versions such as Llama 3.2 from Meta, have been be shrunk to as small as 2GB. This is accomplished through sophisticated mathematical techniques that act like data compression, removing redundant information and simplifying the model's structure without significant loss of performance.

Large Language Models (LLMs) are just one category of AI and although out of scope for this document, some other common types of AI are listed below.

AI Types	Description
Predictive AI	Analyses historical data to forecast future outcomes, crucial for everything from weather reports to stock market trends.
Computer Vision	Enables machines to interpret and understand visual information, forming the backbone of self-driving cars and medical imaging analysis.
Task-Oriented AI (RPA)	Automates specific, repetitive processes, such as the robotic arms on an assembly line or software bots handling data entry.

3. Tasks for locally hosted AI Large Language Models?

The videos linked demonstrate the general setup and capabilities that enable these specific functions.

1. **Process and Summarise Documents (Contracts, Policies, Procedures, Reports, Technical Info)**
    - Load company or 3rd party documents and ask questions in natural language.
    - **Demo:** Feed Documents to a Local Large Language Model (first 2mins are different topic)
        - Video demonstrates explains the two options for feeding documents to LLMs – via the Context Window and Retrieval Augmented Generation (RAG).
        - Web sites can also be scanned for information.
2. **Drafting Documents on any topic.**
    - Example: Try "*We are a small business which holds some customer data. Draft a 20 point checklist for an upcoming GDPR assessment.*" on ChatGPT, Google Gemini, or Ollama running local LLMs on a Desktop or Laptop PC – including using voice rather than the keyboard.
3. **Brainstorm Ideas and Strategies**
    - Use any LLM as a creative partner to explore new ideas or strategies.
4. **Create Meeting Agendas and Summaries**
    - Provide a few bullet points and have the LLM flesh out a full meeting agenda or paste raw notes and ask for a clean summary.
5. **Microsoft 365 Copilot – Excel, Outlook, PowerPoint, Teams, Word**
    - **Overview:** Microsoft Copilot
    - **Excel:** Copilot Excel Tutorial
    - **Word:** Copilot in Word | Transform a document
    - **Outlook and Teams:** Outlook and Teams: Enhance Communications

4. Self Hosting – Benefits, Hardware and Security.

Self-hosted LLMs will not outperform premium AI services from Meta, Google, OpenAI (ChatGPT), Microsoft and others for quality. However, provided each self-hosted solution is 'good enough' for its task(s), being able to state that no client or confidential documents/information passes outside of your organisation is valuable.

Costs need not be high for small organisations or personal use.

- For individual users, reasonably modern Desktop or Laptop computers with at least 8GB of system RAM, will run Llama 3.2 from Meta and some other small LLMs listed in Section 5 below. Ollama, LMStudio and AnythingLLM are easy to use and good for getting started.
- If performance is an issue on a particular PC, a £200 Nvidia Graphics Card (GPU) with at least 6GB will work for models whose 'physical size' is smaller than the GPU memory size.
- For teams and departments, LLMs can be securely self-hosted on Servers using Open Source software e.g., Ollama, Open-WebUI, Searxng and Stable Diffusion (for multi-media), running under Docker.
- For Servers, graphics cards will be required but again, consumer grade Nvidia Graphics cards such as the RTX 3060 12GB and above, will be sufficient provided there is sufficient total GPU memory to comfortably hold all the LLMs that are being used in any given moment.

- **Selection of free to download and use Large Language Models.**

The models listed below can all be downloaded from reputable community libraries such as Ollama Library and Hugging Face.

Model Name	Size (Parameters)	Physical Size (GB)	Producer	Generates Images/Video	Notes
Qwen 1.5 7B	7B	4.6	Alibaba Cloud	No	Strong multilingual model, excellent for global business communications and content localization.
Qwen 1.5 14B	14B	8.3	Alibaba Cloud	No	Larger version with a 32k context window, suitable for analysing long documents and reports.
AnimateDiff v3	~1.1B	~5.0	ByteDance / Community	Yes - Video	A motion module used with image models (like SD 1.5) to create text-to-video animations. Very VRAM-efficient.
Gemma 2 9B	9B	5.4	Google	No	A powerful and versatile model from Google, suitable for a wide range of business tasks.
Gemma 7B	7B	4.8	Google	No	Lightweight and efficient; great for sentiment analysis, text summarization, and customer support chatbots.
Llama 3 8B	8B	4.7	Meta	No	Excellent general-purpose model with top-tier performance in reasoning, coding, and instruction following.
Code Llama 13B	13B	7.3	Meta	No	A specialized coding assistant that can significantly speed up software development and debugging cycles.
Phi-3 Medium	14B	7.9	Microsoft	No	A powerful and versatile model that offers a great balance of performance and efficiency for general tasks.
Phi-3 Mini	3.8B	2.3	Microsoft	No	A very lightweight yet capable model designed to run on-device, ideal for mobile and edge computing apps.
Mistral 7B	7B	4.1	Mistral AI	No	High-performing and efficient, known for strong reasoning and language understanding.
Yi 1.5 9B	9B	5.2	01.AI	No	A high-performing open-source model with strong bilingual (English/Chinese) capabilities.
Playground v2.5	2.6B	6.9	Playground AI	Yes - Image	Creates aesthetically pleasing images. Perfect for marketing, social media, and product mockups.
Stable Diffusion 1.5	0.9B	4.3	Stability AI	Yes - Image	A fast and lightweight model for rapid image generation. Great for iterative design and concept art.
Stable Diffusion XL	2.6B	6.9	Stability AI	Yes - Image	The flagship image model. Produces high-resolution, detailed images in a wide variety of styles.
Stable Video Diffusion	2.2B	5.2	Stability AI	Yes - Video	Generates short video clips from a starting image. Useful for creating dynamic ads, GIFs, and animated content.